Q.24 What is effect of superheating the suction vapour on the performance of a vapour compression system?

Q.25 Explain reverse Carnot cycle refrigerator.

Q.26 Write the properties of R-134a.

Q.27 Write uses of R-22.

Q.28 Write the advantages of solar power refrigeration system over vapour compression refrigeration system.

Q.29 Explain semi-hermetically reciprocating compressor.

Q.30 Explain flooded type evaporator.

Q.31 Explain thermostatic expansion valve.

Q.32 Write a short note on low-pressure and high-pressure cut out switches.

Q.33 Write the disadvantages of window type air conditioner.

Q.34 Explain the importance of psychrometric chart.

Q.35 Explain vapour compression refrigeration system.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 Explain domestic Electrolux refrigeration system with the help of neat sketch.

Q.37 Explain the Humidification and dehumidification psychrometric process in detail

Q.38 Explain the working of central air-conditioner with the help of neat sketch. Write the advantages and disadvantages of it.

Roll No. ....

5th Sem / Mech. Engg. MSIL

Sub : Refrigeration & Air Conditioning

Time : 3Hrs. M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 The ratio of heat extracted in the refrigerator to the work done on the refrigerator is called

a) COP of heat pump
b) COP of heat engine
c) Refrigerating efficiency
d) COP of refrigerator

Q.2 For a vapour compression refrigeration system to have high COP, it should have

a) Higher suction pressure
b) Higher evaporator temperature
c) Lower condenser temperature
d) All of these

Q.3 Which of the following refrigerant has the lowest boiling point?

a) Ammonia b) Carbon dioxide
c) R-134a d) R-12

Q.4 What is the pressure at a refrigerators inlet known as?
- a) Back Pressure
- b) Critical Pressure
- c) Discharge Pressure
- d) Suction Pressure

Q.5 In a vapour absorption system the energy supplied as compared to refrigerating effect obtained is about
- a) 50% b) 75%
- c) 100% d) 150%

Q.6 Which of the following refers to the term C.O.P. of refrigerator?
- a) Cooling for performance
- b) Coefficient of performance
- c) Capacity of performance
- d) Coefficient of plant

Q.7 Vapour absorption system makes use of
- a) Kinetic energy b) Potential energy
- c) Mechanical energy d) Het energy

Q.8 The natural convention air cooled condensers are used in
- a) Domestic refrigerators
- b) Water coolers
- c) Room air conditioners
- d) All of these

Q.9 During sensible heating of air, the humidity ratio
- a) Increases b) Decreases
- c) Remain constant d) None of these

Q.10 A split air conditioner does not have provision for
- a) Re-circulation of air b) Ventilation of air
- c) Cleaning of Air d) Exhaust of air

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 The value of COP should be greater than ____.

Q.12 A refrigerator is a reverse of heat engine. (True/Flase)

Q.13 Define the refrigeration effect.

Q.14 Write the function of suction line in simple vapour compression.

Q.15 The sub-cooling of refrigerant increases the COP. (True/False)

Q.16 Define primary refrigerants.

Q.17 The COP of solar power refrigeration system is usually less than ______.

Q.18 A cooling tower cools the water to the web bulb temperature of air. (True/False)

Q.19 What is overload protector?

Q.20 The process in which air is cooled without changing its moisture content is called_____.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Write the properties of an Ideal refrigerant.

Q.22 Explain the Evaporative refrigeration

Q.23 Write the various advantages of air refrigeration system.